

Chromatic Completeness and the Independence of Geometric Obstruction

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We establish a strict logical separation between two distinct phenomena in the theory of orthogonality hypergraphs: *chromatic completeness*—the ability to assign a globally consistent spectrum to maximal observables across all contexts—and *geometric coordinatizability*—the existence of a faithful orthogonal representation (FOR). While Shekarriz and Svozil observed that a chromatic number $\chi > n$ obstructs chromatic completeness in dimension n , this condition does not constrain the existence of an FOR. We show that spectral assignments constitute an additional layer of structure on top of geometric realizations; consequently, chromatic obstructions constrain the spectral labeling, not the underlying geometry. The Yu–Oh configuration ($\chi = 4$) provides an explicit counterexample to any implication from chromatic number to geometric non-coordinatizability. Conversely, to demonstrate that geometric obstruction is an independent phenomenon, we provide an elementary algebraic proof that Greechie’s G_{32} hypergraph admits no faithful orthogonal representation in $\mathbb{C}^3$, a result derived solely from the incidence geometry of its rays without reference to its chromatic properties.

I. INTRODUCTION

An orthogonality hypergraph $H = (V, E)$ consists of a vertex set V and a collection E of hyperedges (contexts), each comprising n mutually orthogonal rank-one projectors in a prospective Hilbert space $\mathcal{H}^n$. A *faithful orthogonal representation* (FOR) is a geometric realization: an assignment of distinct rays $[v_i] \in \mathbb{P}(\mathcal{H}^n)$ to vertices such that orthogonality within every hyperedge is preserved. The existence of such representations is the central question in the quantum logic of the Kochen–Specker theorem [1].

Recent work by Shekarriz and Svozil [2, 3] introduced *chromatic contextuality*, observing that hypergraphs with chromatic number $\chi(H) > n$ cannot support representations where every context carries the same fixed spectrum. This has led to a subtle conflation in the literature: the impression that $\chi > n$ obstructs the existence of an FOR *tout court*.

The primary result of this paper is that such a conflation mistakes a property of the *spectral labeling* for a property of the *underlying geometry*. We formalize the notion of *chromatic completeness* (Definition 2) as the requirement that an FOR admit a global labeling of vertices by eigenvalues such that each context realizes the same spectrum. This is a constraint on the *spectral assignment*, not on the *geometry*. We prove:

Proposition 1 (Independence). *The existence of an FOR in $\mathcal{H}^n$ is independent of the chromatic number $\chi(H)$. Specifically:*

- (i) *If $\chi(H) \leq n$, an FOR may or may not exist;*
- (ii) *If $\chi(H) > n$, an FOR may or may not exist.*

The first clause is trivial; the second is established by the Yu–Oh configuration [5], which has $\chi = 4$ yet possesses a well-known FOR in $\mathbb{R}^3$. Thus chromatic bounds

govern only the existence of *chromatically complete* representations. The proof that G_{32} is nevertheless non-coordinatizable despite having the same chromatic number shows that geometric obstruction is a separate phenomenon requiring direct geometric analysis.

To close the logical loop, we address the converse question: if chromatic properties do not forbid geometric realizability, what does? We treat Greechie’s G_{32} hypergraph [4]—which has $\chi = 4$ —as a case study in pure geometric obstruction. By an explicit algebraic argument, we demonstrate that G_{32} admits no faithful orthogonal representation in $\mathbb{C}^3$. The obstruction arises from the rigid incidence geometry of the configuration, manifesting as an algebraic identity that forces distinct vertices to occupy identical rays. This establishes that G_{32} is non-coordinatizable for reasons entirely distinct from its chromatic number.

II. SPECTRAL ASSIGNMENTS AND GEOMETRIC REALIZABILITY

To clarify the relationship between geometry and colorings, we introduce the following structure. Let $\mathcal{R}(H, \mathcal{H}^n)$ denote the set of faithful orthogonal representations of H in $\mathcal{H}^n$ (the *realization space*). For a fixed representation $\rho \in \mathcal{R}$, each hyperedge $e \in E$ corresponds to an orthonormal basis $\{[v_i]\}_{i \in e}$. A maximal observable for this context is any operator $A_e = \sum_{i \in e} \lambda_i P_i$ with distinct eigenvalues.

Definition 2 (Chromatic Completeness). *A representation ρ is chromatically complete if there exists a single set $\Sigma = \{\lambda_1, \dots, \lambda_n\}$ of distinct scalars and a function $c: V \rightarrow \Sigma$ such that for every context $e = \{v_{i_1}, \dots, v_{i_n}\}$, the restriction $c|_e$ is a bijection onto Σ .*

In other words, chromatic completeness requires the eigenvalue assignment to be a proper n -coloring of the hypergraph. The observation of Shekarriz and Svozil is

that $\chi(H) > n$ implies no chromatically complete representation exists. This is correct, but it constrains only the *spectral labeling* associated with ρ , not the geometry.

Indeed, fix a geometric realization ρ . The set of possible spectral assignments for a single context e is the symmetric group S_n (permuting the n eigenvalues onto the n rays). A chromatically complete representation requires a globally consistent choice of eigenvalue permutations across all contexts—equivalently, a proper n -coloring. The Yu–Oh configuration demonstrates that such a globally consistent assignment may not exist (no 3-coloring exists) while the geometric realization space $\mathcal{R}$ is nonetheless nonempty.

The physical irrelevance of chromatic completeness follows from the functional calculus. Two maximal operators sharing the same spectral projections differ only by a relabeling of their outcomes.

Lemma 3 (Spectral Equivalence). *Let*

$$A = \sum_{i=1}^n a_i P_i$$

be a maximal observable on $\mathcal{H}^n$, where the spectral projections $\{P_i\}$ are one-dimensional and the eigenvalues a_i are distinct. Then every maximal observable sharing the same spectral projections is of the form

$$B = f(A),$$

where f is injective on the spectrum of A . Conversely, every injective function on the spectrum of A produces a maximal observable with the same eigenspaces.

Proof. Suppose

$$B = \sum_{i=1}^n b_i P_i$$

shares the same spectral projections as A . Since A and B possess the same eigenspaces, they commute and are simultaneously diagonalizable. The spectral theorem therefore implies that B is obtained by applying a function to the eigenvalues of A .

Define

$$f(a_i) = b_i, \quad i = 1, \dots, n.$$

Because both observables are maximal (non-degenerate), the numbers b_i are distinct, so f is injective on the finite spectrum of A .

An explicit realization of f is given by the Lagrange interpolation polynomial

$$f(x) = \sum_{i=1}^n b_i \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - a_j}{a_i - a_j},$$

which satisfies $f(a_i) = b_i$ for every eigenvalue. Consequently, $B = f(A)$.

Conversely, any injective map on the spectrum preserves the distinctness of the eigenvalues while leaving the spectral projections unchanged, producing another maximal observable with the same eigenspaces. $\square$

III. THE YU–OH CONFIGURATION: GEOMETRIC REALIZABILITY WITHOUT CHROMATIC COMPLETENESS

The separation asserted in Proposition 1(ii) is instantiated by the Yu–Oh hypergraph H_{YO} which has a vertex set $\{1, \dots, 25\}$ and 16 blocks [5]:

$$\begin{aligned} &\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}, \{2, 5, 8\}, \\ &\{9, 10, 23\}, \{4, 11, 24\}, \{3, 12, 25\}, \{1, 13, 20\}, \\ &\{6, 13, 22\}, \{7, 13, 21\}, \{3, 11, 14\}, \{7, 11, 15\}, \\ &\{9, 12, 16\}, \{6, 12, 17\}, \{4, 10, 18\}, \{1, 10, 19\}. \end{aligned}$$

It is 3-uniform (all contexts contain 3 vertices) and has chromatic number $\chi(H_{YO}) = 4$.

If chromatic number obstructed geometric realizability, H_{YO} would admit no FOR in $\mathbb{R}^3$. However, an explicit coordinatization exists [5]. These 13 distinct rays realize all 9 contexts of H_{YO} with mutual orthogonality preserved.

The non-existence of a 3-coloring implies that no choice of maximal operators on this geometry can share a common spectrum. But by Lemma 3, we may simply assign independent spectra to distinct contexts. For instance, context C_1 may carry eigenvalues $\{0, 1, 2\}$ while an overlapping context C_2 carries $\{0, \pi, e\}$; the shared projector receives different numerical labels in each context, related by an injective remapping. The Yu–Oh configuration thus demonstrates that the chromatic obstruction is confined to the spectral labeling and does not affect the existence of the geometric realization.

IV. GEOMETRIC OBSTRUCTION IN G_{32}

Having established that chromatic contextuality does not preclude geometric realizability, we now demonstrate that geometric obstruction is nevertheless a genuine and distinct phenomenon. We prove by elementary algebraic methods that Greechie’s diagram G_{32} admits no faithful orthogonal representation in $\mathbb{C}^3$.

A. The hypergraph

The Greechie diagram or hypergraph G_{32} has vertex set $\{1, \dots, 15\}$ and 10 blocks (triples of mutually orthogonal rays):

$$\begin{aligned} &\{1, 2, 3\}, \{3, 4, 5\}, \{5, 6, 7\}, \{7, 8, 9\}, \{9, 10, 11\}, \\ &\{11, 12, 1\}, \{4, 10, 13\}, \{6, 12, 14\}, \{8, 2, 15\}, \{13, 14, 15\}. \end{aligned}$$

B. Main theorem

Theorem 4. *The hypergraph G_{32} admits no faithful orthogonal representation in $\mathbb{C}^3$.*

Proof. Assume an FOR exists. We derive two independent algebraic constraints on the transition amplitudes between certain vertices, which together force a contradiction with faithfulness.

a. Step 1: Gauge fixing. Since $\{13, 14, 15\}$ is a block, its vertices form an orthonormal basis. By a global unitary transformation, we fix:

$$v_{13} = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_{14} = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_{15} = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

b. Step 2: Spoke parametrization. The three spoke contexts $\{4, 10, 13\}$, $\{6, 12, 14\}$, and $\{8, 2, 15\}$ connect the central triangle to the outer cycle. Each spoke forces two vectors to lie in a coordinate two-plane, reducing the problem to three pairs of complex parameters.

- *Block $\{4, 10, 13\}$:* $v_4, v_{10} \perp e_1$, so they lie in $\text{span}(e_2, e_3)$. Write

$$v_4 = \begin{pmatrix} 0 \\ a \\ b \end{pmatrix}, \quad |a|^2 + |b|^2 = 1,$$

and take the unique orthogonal ray in this plane as

$$v_{10} = \begin{pmatrix} 0 \\ -\bar{b} \\ \bar{a} \end{pmatrix}.$$

- *Block $\{6, 12, 14\}$:* $v_6, v_{12} \perp e_2$, lying in $\text{span}(e_1, e_3)$:

$$v_6 = \begin{pmatrix} c \\ 0 \\ d \end{pmatrix}, \quad v_{12} = \begin{pmatrix} -\bar{d} \\ 0 \\ \bar{c} \end{pmatrix}, \quad |c|^2 + |d|^2 = 1.$$

- *Block $\{8, 2, 15\}$:* $v_8, v_2 \perp e_3$, lying in $\text{span}(e_1, e_2)$:

$$v_2 = \begin{pmatrix} p \\ q \\ 0 \end{pmatrix}, \quad v_8 = \begin{pmatrix} -\bar{q} \\ \bar{p} \\ 0 \end{pmatrix}, \quad |p|^2 + |q|^2 = 1.$$

c. Step 3: A projection identity. The key technical ingredient is a simple consequence of the resolution of the identity.

Lemma 5. *If $\{x, y, z\}$ is an orthonormal basis of $\mathbb{C}^3$, and $u \perp x$, $w \perp z$, then*

$$\langle u, w \rangle = \langle u, y \rangle \langle y, w \rangle. \quad (1)$$

Proof. Expanding

$$w = \langle x, w \rangle x + \langle y, w \rangle y + \langle z, w \rangle z,$$

the last coefficient vanishes because $w \perp z$, while taking the inner product with u eliminates the first because $u \perp x$. Only the middle term survives. $\square$

d. Step 4: Two forced transition-amplitude identities. The outer twelve-cycle provides two independent propagation paths. Applying the projection identity (1) along opposite halves of the cycle yields two factorizations of transition amplitudes between the spoke vectors.

Identity A: Using block $\{1, 2, 3\}$ as an orthonormal basis $\{v_3, v_2, v_1\}$ (with $v_3 \perp v_4$ from block $\{3, 4, 5\}$ and $v_1 \perp v_{12}$ from block $\{11, 12, 1\}$), Eq. (1) gives:

$$\langle v_4, v_{12} \rangle = \langle v_4, v_2 \rangle \langle v_2, v_{12} \rangle. \quad (2)$$

Identity B: Using block $\{7, 8, 9\}$ as an orthonormal basis $\{v_9, v_8, v_7\}$ (with $v_9 \perp v_{10}$ from block $\{9, 10, 11\}$ and $v_7 \perp v_6$ from block $\{5, 6, 7\}$):

$$\langle v_{10}, v_6 \rangle = \langle v_{10}, v_8 \rangle \langle v_8, v_6 \rangle. \quad (3)$$

e. Step 5: Algebraic evaluation. Substituting the explicit parametrizations into the two identities produces two independent algebraic constraints relating the six free parameters:

$$\begin{aligned} \langle v_4, v_{12} \rangle &= \bar{b}\bar{c}, & \langle v_4, v_2 \rangle &= \bar{a}q, & \langle v_2, v_{12} \rangle &= -\bar{p}\bar{d}, \\ \langle v_{10}, v_6 \rangle &= ad, & \langle v_{10}, v_8 \rangle &= -b\bar{p}, & \langle v_8, v_6 \rangle &= -qc. \end{aligned}$$

From (2) we obtain $\bar{b}\bar{c} = -\bar{a}\bar{d}q\bar{p}$, or upon complex conjugation:

$$bc = -ad\bar{q}p. \quad (4)$$

From (3) we obtain:

$$ad = bc\bar{p}q. \quad (5)$$

f. Step 6: Derivation of the contradiction. Substituting (4) into (5) yields:

$$ad = (-ad\bar{q}p)\bar{p}q = -ad|p|^2|q|^2.$$

Since $|p|^2 + |q|^2 = 1$, the factor $1 + |p|^2|q|^2$ is strictly positive, forcing $ad = 0$.

If $a = 0$, then $|b| = 1$ and $v_4 = be_3$, implying $[v_4] = [v_{15}]$, contradicting the distinctness of atoms 4 and 15.

If $d = 0$, then $|c| = 1$ and $v_6 = ce_1$, implying $[v_6] = [v_{13}]$, contradicting the distinctness of atoms 6 and 13.

Thus no faithful representation exists. $\square$

C. Remark on the nature of the obstruction

The contradiction arises from the algebraic rigidity of the G_{32} incidence structure, specifically the spoke geometry connecting the outer twelve-cycle to the central triangle $\{13, 14, 15\}$. This is a purely projective-geometric phenomenon: the constraints on inner products propagate around the cycle to force a ray collision. It bears no relation to the chromatic number $\chi = 4$; indeed, the proof never invokes coloring arguments. The obstruction is therefore strictly geometric.

V. CONCLUSION

We have shown that the existence of a faithful orthogonal representation is logically independent of the chromatic number. The condition $\chi > n$ obstructs only the *chromatically complete* representations—those admitting a global spectral labeling—not the geometric realization itself. The Yu–Oh configuration provides the definitive counterexample: $\chi = 4$ in $\mathbb{R}^3$, yet a faithful representation exists. Conversely, G_{32} demonstrates that geometric obstruction is a separate, rigorously demonstrable phe-

nomenon.

In assessing whether a hypergraph admits an FOR, one must therefore look to the incidence geometry and algebraic dependencies of its rays, not to the combinatorics of its colorings. Chromatic contextuality concerns the global consistency of measurement labels, whereas geometric coordinatizability concerns the realizability of measurement directions. These are logically distinct aspects of quantum contextuality and should not be conflated.

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